

INDIAN STANDARDS

DRAFT FOR COMMENTS ONLY

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भारतीय मानक मसौदा
चमड़ा - शू अपर के लिए ग्लेज्ड किड — विशिष्टि
(IS 576 का तीसरा पुनरीक्षण)

Draft Indian Standard
Leather - Glazed Kid for Shoe Upper – Specification
(Third Revision of IS 576)

(ICS 59.140.20)

Leather, Tanning Materials and Allied Products Sectional
Committee, CHD 17

Last Date for Comments: **As per covering letter**

FOREWORD

(Formal clause will be added later)

This standard was first published in 1954. The first revision of this standard was published in 1975 incorporating additional physical requirements, namely, thickness, elongation at break, flexural endurance (Bally), colour fastness to rubbing, and chemical requirements, such as, hide substance and total ash. The second revision of this standard was published in 1989 by incorporating requirements on distension at grain crack and load at grain crack to consider the footwear industry's needs to ensure better leather performance during lasting operation.

During this third revision the following changes have been incorporated:

- a) Restricted/critical chemical substances requirements have been added;
- b) Colour fastness to water spotting and water vapour coefficient, mg/cm² have been added; and
- c) Amendment No. 1 May 1998 has been incorporated.

In reporting the result of a test or analysis made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS 2: 2022. 'Rules for rounding off numerical values (second revision)'.

1 SCOPE

1.1 This standard prescribes the requirements and methods of sampling and test for glazed kid leather for shoe upper.

2 REFERENCES

The standards listed in Annex A contain provisions which through reference in this text, constitute provisions of

this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent edition of these standards.

3 TERMINOLOGY

3.1 For the purpose of this standard, definitions given in IS 1640 shall apply.

4 REQUIREMENTS

4.1 Raw Material

The material shall be wet salted/green goat skins of medium size with good substance. The skins shall be free from grain and flesh defects.

4.2 Tanning

The leather may be chrome-tanned, combination-tanned, or other tanning as agreed upon by the buyer and purchaser. The leather tanned with basic chromium salts shall not shrink when tested in the blue stage in boiling water as prescribed in LP: 10 of IS 5914: 1970.

4.3 Post tanning

The material shall be shaved such that the final finished leather is in accordance with the thickness (± 0.1 mm) as agreed to between the purchaser and the supplier. At the end of tanning operation, before drying, suitable fungicides shall be incorporated of leather in requisite proportions. Fungicides used to promote mildew resistance in the leather shall be effective and harmless to skin or hide.

4.4 Dyeing

The leather shall be dyed to an even shade and the dye shall penetrate throughout the grain layer or as agreed between buyer and purchaser.

4.5 Finish

Glazed kid leathers shall be finished using traditional albumen-based seasons, pigment finishes, or a combination of both. Synthetic chemical formulations capable of rendering properties similar to those of albumen-based season may also be used. The finish shall be glazed to a highly glossy and plain surface.

4.6 Chemical Requirements

Glazed kid leather shall conform to the chemicals of quality and performance requirements as specified in Table 1 when tested by the relevant test methods as referred.

Table 1 Chemical Requirements for Glazed Kid Leather for Shoe Upper

(Clause 4.6)

Sl. No	Characteristics	Requirements	Method of Test, Ref to
(1)	(2)	(3)	(4)
i)	Total ash, percent by mass, <i>Max</i> (over Cr ₂ O ₃ content)	2.5	IS 582 (Part 3)
ii)	Solvent extractable substances, percent by mass	2.5 to 7.5	IS 582 (Part 14)
iii)	Chromium (as Cr ₂ O ₃) (if chrome tanned or combination tanned with chromium), percent by mass, <i>Min</i>	2.0	IS 582 (Part 10/Sec 1)
iv)	Hide substance, percent by mass, <i>Min</i>	65.0	IS 582 (Part 12)
v)	Water soluble matter, percent by mass, <i>Max</i>	5.0	IS 582 (Part 2)
vi)	pH of water solubles, <i>Min</i>	3.5	IS 582 (Part 9)
vii)	Differential number (shall not exceed)	0.7	

Sl. No	Characteristics	Requirements	Method of Test, Ref to
NOTE — Calculation of requirements for characteristics (i) to (v) are on zero percent moisture basis. The moisture content shall be determined in accordance with LC: 2 of IS 582.			

4.7 Restricted/critical chemical substances requirements

Glazed kid leather shall conform to the list of restricted/critical chemical substances of Category 1 & 2 prescribed under the *clause* 4.1, Table 1 of the IS 17011.

4.8 Thickness

The material shall be shaved such that the final finished leather is in accordance with the thickness (± 0.1 mm) as agreed to between the purchaser and the supplier.

4.9 Physical Requirements

Glazed kid leather shall conform to the physical and quality performance requirements given in Table 2 when tested by the relevant test methods as referred.

Table 2 Physical Requirements for Glazed Kid Leather

(Clause 4.9)

Sl. No.	Characteristics	Requirements	Test Methods
(1)	(2)	(3)	(4)
i)	Tensile strength, N/mm ² , <i>Min.</i>	10	IS 5914 (Part 8)
ii)	Elongation at Break, <i>percent</i>	45-75	
iii)	Tear Strength, N, <i>Min</i>	40	IS 5914 (Part 5/Sec 2)
iv)	Stitch tear strength, N/mm, <i>Min.</i>	40	LP 8 of IS 5914:1970
v)	Grain crack Load, N, <i>Min.</i>	200	IS 14970
vi)	Grain Crack distension, mm, <i>Min.</i>	7.0	
vii)	Colour fastness to rubbing (Vesilc method), <i>Grey Scale Rate</i>		IS 6191 (Part 4)
	a) Dry 150 Rubs	≥3	
	b) Wet 50 Rubs	≥3	
viii)	Colour fastness to water, <i>Grey Scale Rate</i>	≥3	IS 6191 (Part 2)
ix)	Colour fastness to perspiration, <i>Grey Scale Rate</i>	≥3	IS 6191 (Part 6)
x)	Colour fastness to water spotting		IS 6191 (Part 1)
	a) At 30 min	No change or No permanent mark	
	b) At 16 h		
	c) Grey Scale Rate	≥4	
xi)	Flexing endurance		IS 5914 (Part 6/Sec 1)
	a) Dry - 75,000 flexes	No crack	
	b) Wet - 25,000 flexes	No crack and No salt spue	
xii)	Water vapour permeability, mg/cm2/hr	0.8	IS 5914 (Part 12)

Sl. No.	Characteristics	Requirements	Test Methods
xiii)	Water vapour coefficient, mg/cm ²	15	IS 5914 (Part 1)

5 PACKING AND MARKING

5.1 Packing

The leather shall be packed as agreed to between the purchaser and the supplier.

5.2 Marking

5.2.1 The leather shall be legibly marked on the flesh side of each piece with the area in square decimeters.

NOTE — The marking should not cause any disfiguration of the leather or migrate itself to the grain surface of the leather coming in contact with it.

5.2.2 The packages shall be marked with the following:

- a) Name of the manufacturer;
- d) Recognized trade-mark, if any;
- c) Number of pieces of leather;
- d) Total area; and
- e) Month and year of manufacture

5.2.3 BIS Certification Marking

The product(s) conforming to the requirements of this standard may be certified as per the conformity assessment schemes under the provisions of the *Bureau of Indian Standards Act, 2016* and the Rules and Regulations framed thereunder, and the products may be marked with the Standard Mark.

7 SAMPLING AND CRITERIA FOR CONFORMITY

The scale of sampling and criteria for conformity of the material shall be as prescribed in IS 5868 and its respective parts.

ANNEX A
(Clause 2)

LIST OF REFERRED STANDARDS

<i>IS No.</i>	<i>Title</i>
IS 582	Methods of Chemical Testing of Leather
LC 2	Determination of Moisture
Part 2 : 2024	Determination of Water-Soluble Matter, Water-Soluble Inorganic Matter and Water-Soluble Organic Matter (<i>First Revision</i>)
Part 3 : 2017	Determination of sulphated total ash and sulphated water - Insoluble ash (Second Revision)
Part 9: 2022	Determination of pH and difference figure
Part 10/Sec 1:2022	Determination of Chromic Oxide Content Section 1 Quantification by titration
Part 12 : 2022	Determination of nitrogen content and hide substance Titrimetric method
Part 14 : 2022	Determination of Matter Soluble in Dichloromethane and Free Fatty Acid Content
IS 1640:2025	Hides, Skins, and Leather - Vocabulary (Second Revision)
IS 5868 : 2026	Leather - Methods of Sampling Part 3 Specimen and Test Piece Conditioning for Physical and Mechanical Tests (<i>second revision</i>)
IS 5914	Methods of physical testing of leather
LP 8	Determination of double hole stitch Tear strength
LP 10	Measurement of Shrinkage Temperature
Part 1 : 2018	Determination of Water Vapour Absorption
Part 5/Sec 2 :2023	Determination of tear load Section 2 Double edge tear
Part 6/Sec 1 :2023	Determination of flex resistance Section 1 Flexometer method
Part 8 : 2023	Determination Of Tensile Strength And Percentage Elongation
Part 12:2025	Determination of Water Vapour Permeability
IS 6191	Methods of Micro-Biological, Colour Fastness and Microscopical Tests for Leather
Part 1 : 2018	Colour Fastness to Water Spotting
Part 2 : 2017	Colour Fastness to Water
Part 4 : 2024	Colour Fastness to Cycles of to-and-fro Rubbing (<i>First Revision</i>)
Part 6 : 2023	Colour Fastness to Perspiration
IS 14970 : 2025	Leather — Determination of Distension and Strength of Surface (Ball Burst Method)
IS 17011 : 2018	Chemical requirements for footwear and footwear materials